Fiscal Year:	FY 2015	**Task Last Updated:**	FY 09/07/2015
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	BRIC: Mapping Spaceflight-Induced Hypoxic Signaling And Response		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sgilroy@wisc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	608-262-4009
Organization Name:	University of Wisconsin-Madison		
PI Address 1:	Department of Botany		
PI Address 2:	430 Lincoln Dr.		
PI Web Page:			
City:	Madison	**State:**	WI
Zip Code:	53706-1313	**Congressional District:**	2
Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	06/12/2012	**End Date:**	06/11/2015
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
Contact Email:	howard.g.levine@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 6/11/2015 per PI and NSSC information (Ed., 6/20/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX12AK79G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research has capitalized on a rapid turn-around spaceflight opportunity and the Biological Research in Canisters-Petri Dish Fixation Units (BRIC-PDFU) hardware to address how spaceflight affects gene expression related to low oxygen/hypoxic response in the model plant Arabidopsis thaliana. Root zone hypoxia is thought develop in spaceflight as weightlessness leads to a reduction in the buoyancy-driven convection that usually aids in gas exchange around organisms. This in turn leads to the development of oxygen-limiting conditions with adverse effects on plant vigor and yields. Our analysis of Ca2+ signaling responses and associated gene expression patterns in Arabidopsis indicates Ca2+ signaling is linked to hypoxic response in plants. We have identified a suite of Ca2+-dependent proteins that show rapid transcriptional responses to anoxia in both roots and shoots. Mutants in two such genes, aca1-2, and cax2-2 (both Ca2+ transporters) are resistant to hypoxic challenge. These mutants also show disrupted gravitropic response suggesting that Ca2+ signaling may provide a point of cross-talk between the anoxic and gravity response systems. We therefore compared growth and gene expression patterns in cax2-2, and wild-type plants grown on the ground and ion the International Space Station. Seedlings were grown on orbit and fixed in RNAlater for subsequent analysis. Differences in the of gene expression were analyzed post-flight by a combination of RNAseq and qPCR analyses. Comparison to ground-based controls maintained under normal and varying degrees of low oxygen conditions at 1xG, have been used to ask how much of the responses seen on orbit can be explained by development of low oxygen conditions around the plants due to microgravity conditions. Results from this analysis are not only helping advance our understanding of the development of anoxic response in space and on the Earth but also test whether the Ca2+ signaling system provides targets for genetically engineering potential countermeasures to this stress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is addressing how low oxygen sensing operates in plants, targeting the Ca2+-dependent signaling machinery of the plant root. The results from this work are not only providing insight into how low oxygen levels impact plant growth in space but also are contributing to our understanding of similar responses on Earth. The low oxygen sensing/response systems triggered by anoxic conditions in space are also those employed by plants challenged by low oxygen on Earth, such as during flooding of the soil or intense soil microbe respiration that locally depletes oxygen around the plant. Thus, this work aimed at analysis of how Ca2+ signaling is involved in the low oxygen response induced during spaceflight is also revealing a role for this system in this general flooding/low oxygen response system. These results are providing novel molecular targets for potential manipulation to tailor plant growth to varied soil oxygen status, such as promoting flooding tolerance in crop plants.
Task Progress:	FLIGHT: This project has completed the flight, sample recovery, and sample analysis phases. The experimental setup for the BRIC-17 experiment was using surface sterilized seeds of Arabidopsis plated on nutrient agar medium in 60 mm Petri dishes (4 plates of wild-type and 3 each of two anoxia-related CAX2 Ca2+ transporter mutants). The sterile plates were then loaded into Petri Dish Fixation Units (PDFU) and 5 PDFUs plus a HOBO data logger secured inside each of two of the Biological Research in Canister (BRIC) hardware. Germination was delayed by holding the BRIC at 4°C until the hardware was installed on the International Space Station (ISS). Germination was then started by warming the equipment to room temperature. After 8 days at room temperature, the crew injected RNAlater fixative into the PDFUs and the following day, the canisters were placed in the MELFI freezer onboard the ISS at -95°C. Samples were returned using the Dragon capsule in a -32°C Cold Bag, de-integrated, and transported frozen to the University of Wisconsin-Madison for further analysis. All flight procedures were completed successfully. POSTFLIGHT ANALYSIS: Seedling growth was assessed from digital images of the thawed seedlings. A small but significant ($P<0.05$, t-test) redirection from shoot to root growth was observed in the spaceflight wild type plants versus the ground-based controls. This reallocation of growth was missing in the cax2-3 mutant suggesting this line may be resistant to some of the stresses of spaceflight that are responsible for remodeling growth in wild type. TRANSCRIPTIONAL PROFILING: All samples were dissected into root and shoot and all yielded sufficient quantity 12-36 µg/plate and quality of RNA for both quantitative PCR and whole genome transcriptional analyses to monitor gene expression. Isolated RNA was sent to the University of Wisconsin-Madison Biotech Center who performed ribosomal RNA reduction, library construction, and RNAseq-based transcriptome profiling for whole genome gene expression analysis. The complete raw data-set from this RNAseq analysis was mapped to the TAIR version 10 of the Arabidopsis genome using the Tuxedo protocols. These transcriptional patterns were then analyzed for over-represented classes of genes. Analysis of genes induced >5-fold indicated 38 genes were up-regulated in spaceflight in wild-type and in the cax2 mutant plants relative to ground controls, implying a core set of spaceflight-related genes. These genes cover a range of functions from hormone response genes to those related to oxidative stress. In addition, the induction of 210 other genes was suppressed in the cax2 mutants relative to wild type, suggesting these are targets of a CAX2-dependent machinery that is triggered by spaceflight, and so strong candidates for genes that are part of the spaceflight anoxic response that is triggered by the CAX2 protein. In addition, 329 genes were up-regulated in both cax2 mutants but not in wild type plants. As both of these CAX2 mutants are resistant to low oxygen stress on Earth, these up-regulated genes represent strong candidates for potential hypoxia resistance genes. Many of these genes are also elevated in the CAX2 mutant backgrounds in the ground controls but are further increased by the spaceflight environment. Analysis of this subset of genes using the GOrilla gene ontology analysis tool revealed significant enrichment in genes related to hypoxia, oxidative stress, and biotic stress responses.
Bibliography Type:	Description: (Last Updated: 02/02/2026)

Articles in Peer-reviewed Journals	Gilroy S, Suzuki N, Miller G, Choi WG, Toyota M, Devireddy AR, Mittler R. "A tidal wave of signals: Calcium and ROS at the forefront of rapid systemic signaling." Trends Plant Sci. 2014 Oct;19(10):623-30. Epub 2014 Jul 23. http://dx.doi.org/10.1016/j.tplants.2014.06.013 ; PubMed PMID: 25088679 , Oct-2014
Articles in Peer-reviewed Journals	Choi WG, Hilleary R, Swanson S, Kim SH, Gilroy S. "Rapid Long-Distance Electrical and Calcium Signaling in Plants." Annual Reviews in Plant Biology, Submitted as of September 2015. , Sep-2015
Articles in Peer-reviewed Journals	Lourenço TF, Serra TS, Cordeiro AM, Swanson SJ, Gilroy S, Saibo NJ, Oliveira MM. "The rice E3 ubiquitin ligase OsHOS1 modulates the expression of OsRMC, a gene involved in root mechano-sensing, through the interaction with two ERF transcription factors." Plant Physiol. 2015 Nov;169(3):2275-87. Epub 2015 Sep 17. https://doi.org/10.1104/pp.15.01131 ; PubMed PMID: 26381316; PubMed Central PMCID: PMC4634088 [NOTE: reported originally as Epub ahead of print in Sep 2015] , Nov-2015
Articles in Peer-reviewed Journals	Choi WG, Barker RJ, Kim SH, Swanson SJ, Gilroy S. "Variation in the transcriptome of different ecotypes of Arabidopsis thaliana reveals signatures of oxidative stress in plant responses to spaceflight." American Journal of Botany. 2019 Jan;106(1):123-36. https://doi.org/10.1002/ajb2.1223 , Jan-2019